

Report for Geotechnical Investigation CQU Rok Tafe Consolidation

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Distribution List

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1.0 Introduction

1.1 General

This report presents the results of the geotechnical investigation carried out by Soil Surveys Engineering Pty. Limited for the CQU Rok Tafe Consolidation project at the Central Queensland University Rockhampton North Campus.

The investigation was carried out in accordance with Soil Surveys Engineering Pty Limited proposal 1-29688 PR 1.1.

1.2 Proposed Development

It is understood the proposed development includes new buildings and extensions within the Central Queensland University, Rockhampton campus.

The following is proposed:

- New Horticulture building - new single-storey portal framed shed.
- New Trade Building - Two level building with column loads in the 800kN.
- Engineering Shed - Extensions to existing building 21A with column loads of approximately 800kN.
- New lifts to Building 32 and 36.

1.3 Scope of Geotechnical Services

The objective of this study was to identify materials and material properties and groundwater conditions to enable geotechnical assessment of the following: -

- Earthworks recommendations
 - trafficability
 - subgrade preparation
 - compaction standards
 - excavatability
 - suitability of cut for use as fill
 - temporary safe batter angles for cuts
- Foundation recommendations
 - recommended foundation type (high and deep support)
 - bearing capacity
 - skin friction
 - site classification (AS 2870)

- site subsoil class (AS 1170.4-20)
- predicted ground surface movement
- Retaining wall design parameters
- Pavement recommendations
- Construction recommendations (where applicable)
- Site management recommendations

2.0 Site Description

The proposed new structures are located throughout the existing CQU Rockhampton North Campus (refer Figure 1).



Figure 1

The development areas are currently covered with existing structures, carparking and gravel surface roadways. Vegetation consists of generally grassed areas, landscaping and varying sized trees.

Rockhampton Regional Council contour plans indicate the site slopes down from a high point within the south western corner of the site.

Figures 2 to 5 indicate existing site conditions.



Figure 2 - Trade Building Location

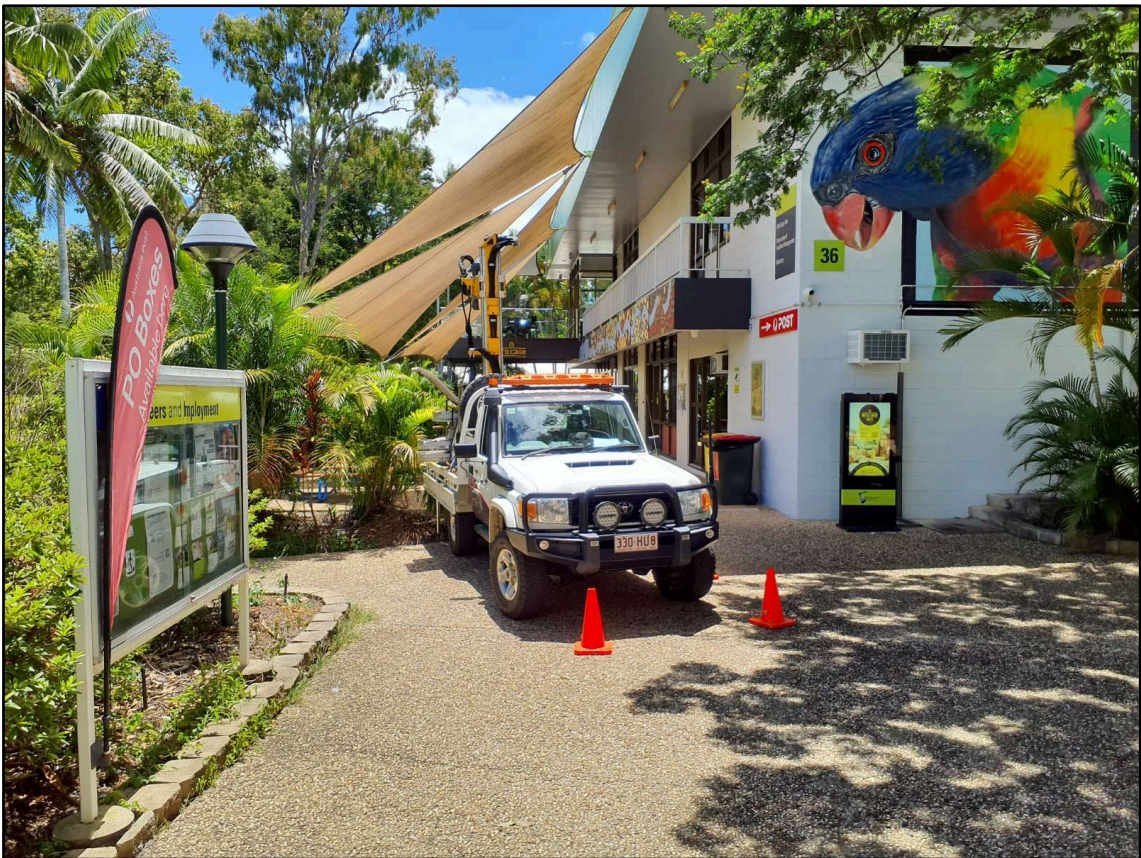


Figure 3 - Building 36, Borehole BH07 Location



Figure 4 - New Horticultural Building Location



Figure 5 - Engineering Shed Extension Location

3.0 Field Investigation

Subsurface conditions at the site were investigated by drilling and sampling 10 (ten) boreholes to depths of between 2.50m and 5.00m, using a 4WD mounted drilling rig. Boreholes BH03, BH04 and BH05 were terminated at 'TC' bit refusal depth. In addition, dynamic cone penetrometer (DCP) tests were carried out adjacent to each borehole.

Borehole locations were nominated by WSP.

The field work was carried out on the 9th and 10th of December 2025.

The soil classification descriptions, field and laboratory testing were carried out in general accordance with the following Australian Standards: -

- AS 1726-2017 - Geotechnical Site Investigations
- AS 1289 - Methods of Testing Soils for Engineering Purposes

A description of the investigation method, borehole records and a site plan showing the location of the boreholes are included in the Appendices.

Borehole coordinates were recorded using a handheld GPS device, with accuracy consistent with such devices.

4.0 Geotechnical Model

4.1 Local Geology

Published geological maps for the local area (Geological Survey of Queensland 1:250 000 series - Rockhampton) indicate that the site is underlain by alluvium soils overlying the Rockhampton Group.

4.2 Subsurface Profile

The subsurface profile intersected during the drilling program consisted of: -

- Fill - Encountered in each borehole (except BH05) to depths of between 0.18m and 0.80m. The fill comprised predominantly Silty Gravel (pavement) and Sandy Clay.

In the absence of documentation to confirm that the filling was placed in controlled conditions in accordance with AS 3798-2007 the filling must be considered 'uncontrolled'.

- Natural Soils - High plasticity Silty Clay (CH) was encountered below the fill (if encountered), which graded into low to medium plastic Sandy Clay (CL/CI) with depth. The consistency of the Clay was very stiff and hard.

Thin layers of Clayey Gravel were intersected within the Clay stratum at the location of Boreholes BH02, BH03 and BH10.

- Weathered Rock - Weathered Siltstone was encountered in the majority of the boreholes. Assessed as extremely weathered (XW) and distinctly weathered (DW), low strength with increased depth.

Boreholes BH03, BH04 and BH05 refused (drill rig refusal) within the DW stratum. Therefore, stronger rock is expected to be encountered below refusal depth.

A summary of the subsurface profile is presented in Table 1 with detailed borehole records included in Appendix B.

Table 1 Summary of Subsurface Profile

Location	Fill (m)	Clay (m)	Clayey Gravel	Weathered Rock (m)		Termination Depth (m)
				XW	DW	
BH01	0.00-0.08	0.08-2.40 3.50-TD	NE	2.40-3.50	NE	4.50
BH02	0.00-0.70	0.70-2.80 3.30-4.00	2.80-3.30	4.00-TD	NE	4.50
BH03	0.00-0.20	0.20-1.10	1.60-1.60	1.60-2.40	2.40-TD	2.50 ³
BH04	0.00-0.18	0.18-1.60	NE	1.60-2.50	2.50-TD	2.70 ³
BH05	NE	0.00-2.60	NE	2.60-3.40	3.40-TD	3.60 ³
BH06	0.00-0.50	0.50-TD	NE	NE	NE	4.50
BH07	0.00-0.80	0.80-4.30	NE	4.30-TD	NE	5.00
BH08	0.00-0.35	0.35-TD	NE	NE	NE	4.50
BH09	0.00-0.80	0.80-3.80	NE	3.80-TD	NE	4.50
BH10	0.00-0.20	0.20-0.60	0.60-1.80	0.80-TD	NE	4.50

Notes:-

1. NE = Not Encountered; TD = Termination Depth.
2. All depths below existing ground level at date of drilling.
3. 'TC' bit refusal.
4. AS 1726-2017 'Geotechnical Site Investigations' requires that XW rock be described using soil terms

4.3 Groundwater

Groundwater was not encountered at borehole locations at the time of the investigation. Whilst groundwater was not encountered in the boreholes, it should be noted that groundwater may be present; please note that the period of time available (during the drilling of boreholes) is relatively short, and hence, the opportunity to observe the presence (or otherwise) of groundwater is limited.

However seepage may be encountered, particularly within the fill material and at the fill/natural and soil/rock interfaces, following rain. It should also be noted that groundwater levels can vary both seasonally and with prevailing weather conditions and, if construction is to be undertaken at a

“significant” time following this investigation, or following ‘wet’ weather, then it would be prudent to confirm groundwater levels prior to construction.

4.4 Laboratory Testing

Laboratory testing was carried out on selected samples retrieved from the site investigation program and was directed towards assessing the reactivity and subgrade characteristics of the subsurface material.

Laboratory testing included:

- Shrink/Swell Index - to assess the reactivity of the subsurface material.
- Swell Pressure - to assess the potential uplift forces resulting from moisture variation within the upper level clays.
- California Bearing Ratio - to assess the subgrade characteristics of the subsurface material under soaked conditions.

The results of the laboratory testing are summarised in Table 2 with certificates contained in Appendix C.

Table 2 Laboratory Test Results

Location	Depth (m)	Results
BH01	0.60-0.90	Iss = 3.8%; Swell Pressure = 120kPa
BH02	1.00-1.30	Iss = 3.5%; Swell Pressure = 250kPa
BH05	1.00-1.25	Iss = 3.9%; Swell Pressure = 60kPa
BH06	0.80-1.10	Iss = 4.0%; Swell Pressure = 120kPa
BH09	1.00-1.30	Iss = 1.3%; Swell Pressure = 120kPa
BH04	0.18-0.70	Soaked CBR = 2.0%
BH06	0.50-1.00	Soaked CBR = 3.5%
BH08	0.30-1.00	Soaked CBR = 3.0%

5.0 Engineering Assessment

5.1 Trafficability

Trafficability conditions were considered to be good during the investigation however some problems may be anticipated during wet weather for even light weight 4WD vehicles.

The soils on site are sensitive to repetitive vehicle loading and water (i.e. they will lose strength through repetitive vehicle loading or if they become overly moist or wet). Further, seepage may also result in a subsequent loss of strength. This may limit trafficability and create difficulties for earthworks operations. This situation would be more pronounced if rainfall followed initial clearing, stripping and grubbing.

Problems may also arise from disturbance of the upper level soil fabric with removal of vegetation and existing structures. Depressions could be formed resulting in water traps and potential softening of adjacent and underlying soils.

It is recommended that after stripping, clearing and grubbing, the exposed surface in the construction area be proof rolled (where appropriate) to assist in identifying weak areas and to improve trafficability. In areas of cut, proof rolling may be deferred until after the cut operation.

Maintaining adequate drainage conditions is essential. It should be ensured that runoff is diverted away from the construction area to prevent ponding of water. In addition, the construction area should be "sealed" at the completion of each day and in the event of rain.

Potential trafficability problems with this site should not be underestimated. The site will very quickly become untrafficable if appropriate seepage and drainage control measures, along with construction practices appropriate for site conditions, are not maintained.

Nevertheless, the contractor should fully inform themselves of the ground conditions on site prior to commencement of earthworks. This requirement should be explicit in any earthworks specifications or contract.

Working Platforms For Tracked Plant and Heavy Construction Vehicles

The scope of Soil Surveys Engineering's study DOES NOT include the design of a working platform for heavy construction vehicles or heavy tracked plant.

Detailed design of a working platform should be carried out considering the operation of actual machinery proposed to be used. This is particularly important when considering the use of heavy piling rigs and heavy cranes - the piling/crane contractor should be consulted regarding their requirements.

Demolition and Clearing Activities

Extreme care should be exercised during the demolition/clearing phase to ensure that excessive subgrade disturbance is not caused during removal of existing structures, trees, services, etc.

5.2 Earthworks

5.2.1 General

Earthworks are unknown at the time of the investigation. The following provides general recommendations with respect to earthworks and should be reviewed once the earthworks extent has been established.

Earthwork procedures should be carried out in a responsible manner in accordance with AS 3798-2007 "Guidelines on Earthworks for Commercial and Residential Developments" and include the following:-

- Clearing, stripping and grubbing should be carried out in areas subject to earthworks (as trafficability conditions allow). Also all soils containing organic matter should be stripped from the construction area; this material is not considered suitable for use as structural fill.
- Depressions formed by the removal of vegetation, underground elements etc. should have all disturbed weakened soil cleaned out and be backfilled with compacted select material.
- The subgrade should be proof rolled (where appropriate) under the supervision of Soil Surveys Engineering in accordance with methods and equipment as per Clause 5.5 of AS 3798-2007. In areas of cut, proof rolling may be deferred until after the cut operation. Areas demonstrating excessive movement should be treated (dried and recompacted) or removed and replaced with compacted fill. Treatment should be to a standard sufficient so that the subgrade passes proof rolling and that compaction can be achieved in the first layer of fill.
- Any fill material encountered should be considered uncontrolled and requiring treatment (i.e. excavate/condition/replace/compact as required).
- Due to the water sensitive nature of the silty sand/clayey sand soils (fill and natural), **it is recommended that insitu silty sand/clayey sand NOT be used as structural fill.**
- Due to the reactive nature of the insitu clays, it is recommended that insitu clays NOT be used as structural fill. The use of natural clays as fill may result in ground surface movements that exceed design limits.
- Imported select fill (if required) should be cohesive in nature and conform to the following specifications:

○ Liquid Limit	35% maximum
○ Plasticity Index	10% maximum
○ Aggregate Size	75mm maximum
○ Passing 19mm Sieve	80% minimum
○ Passing 0.075mm Sieve	20% minimum
○ Shrink/Swell Index	Maximum of 0.5%
○ Soaked CBR	15% minimum
- Quality testing to confirm imported fill quality should be carried out prior to delivery to site; frequency of testing is subject to assessment by Soil Surveys Engineering, however, a

sufficient number of tests should be carried out so that all parties are confident of material characteristics. Quality test results should be reviewed by Soil Surveys Engineering.

- Contractors should submit details of their proposed fill source (with quality certificates) with their tender submission. The proposed fill source should be reviewed by Soil Surveys Engineering prior to awarding the earthworks contract.
- Pavements gravels should comply with appropriate local council quality specifications for base, sub-base and blanket materials.
- Provided the placement moisture content of the imported fill or select insitu material approximates the optimum moisture content for compaction, suitable compaction should be achievable using typical compaction machinery. The fill material should be compacted in layers not exceeding 250mm, loose thickness. However, layer thicknesses will be dependent on the compaction plant type and size, use of vibration, material type and condition. Final maximum placement layer thicknesses will need to be determined when compaction plant as well as material type and conditions are known. Fill batters should be overfilled and cut-back to design batter angles.
- Guidelines for minimum relative compaction values for insitu soils and imported general fill for the building and pavements are presented in Table 3.

Table 3 Minimum Relative Compaction

Location	Minimum Dry Density Ratio
General Fill/Subgrade	98 ⁽¹⁾
Base Course	98 ⁽²⁾
Subbase	98 ⁽²⁾
Notes: 1. The recommended compactions are percentages of the maximum dry density determined by Australian Standard 1289 (Standard Compaction). 2. For processed pavement gravels, minimum density requirements should be a percentage of maximum dry density in accordance with AS 1289 5.2.1, Modified Compaction.	

- Field density testing should be carried out to check the standard of compaction achieved and the placement moisture content. The frequency and extent of testing should be as per guidelines in AS 3798-2007, Section 8.0.
- Backfilling for service trenches, etc. should use good quality material. The backfill should be placed in uniform layers over the full width of the excavations with the layers not exceeding 200mm thickness, loosely placed using wheeled plant and 100mm loose thickness using hand held vibrating plates. The backfill material should be compacted to the specifications outlined above for insitu or imported cohesive material.

5.2.2 Trenching

Shoring of deep trench excavations is recommended. Suitable precautions to satisfy Health & Safety requirements must be adopted. Construction procedures (i.e. operation of plant, storage of materials, etc.) should also consider the nature of the site soils.

5.2.3 Earthworks Supervision

Engineering supervision of the earthworks operations by Soil Surveys Engineering Pty Limited is recommended.

Following production of AS 3798-2007, the terms "Level 1 and 2 Supervision" have been adopted in earthworks specifications to describe what could also be termed Engineering Supervision. Whilst there is no particular problem with using these terms, there does not seem to be wide agreement as to what Level 1 or 2 Supervision actually means or entails.

Regardless of terminology, it should be made clear in any earthworks specification as to what is actually required in terms of certification.

It is recommended that the following objectives (as a minimum) be incorporated into the earthworks specification:-

- Engineering certification that all general earthworks operations (i.e. stripping, proof rolling of subgrade, subgrade treatment, etc.) have been carried out in accordance with the earthworks specification.
- Engineering certification that fill has been placed and compacted to the required minimum density in accordance with the earthworks specification.
- If required, engineering certification that the controlled fill material is suitable to support a conventional slab on ground floor.
- Engineering certification that the quality of any imported fill complies with the earthworks specification requirements.
- Engineering certification that the stability of cut/fill batters and trenches is adequate.

Engineering certification should be provided by a Registered Professional Engineer of Queensland.

5.2.4 Excavation Characteristics

It is anticipated that excavations will consist of the following:-

- Bulk Cuts - for site stripping and excavation to create building platforms and carpark/driveway areas.
- Trenching - for high level footings and underground services.
- Drilling - for bored pier foundations.

The assessment of excavation characteristics of soil and weathered rock is normally based on the depth of penetration of the drilling rig using the 'TC' bit attachment. These depths are recorded on the Borehole Record Sheets.

When augering, the bits are attached to the flight auger.

The limit of the 'TC' bit is indicative of the limit of excavations using a small dozer in bulk excavations (Cat D 6) or a medium sized backhoe in trenches (Case 580 or M.F. 50).

Generally, below the 'TC' bit limit, large excavators (say 30 tonne) possibly with hydraulic rock breakers are required for excavation.

It is considered that excavations could be carried out to the depths of the 'TC' bit limit using the appropriate plant as indicated above. Below the depth of the 'TC' bit limit, larger plant with specialised attachments, e.g. rock breakers, may be required for the proposed construction. Some budgeting should be allocated for the possibility of encountering zones of stronger rock.

It should be noted that the ability to drill piers in the weathered rock material is not only dependent on material characteristics but also the type (power and size) of the bored pier drilling rig, drilling teeth, size of pier, etc. It is recommended that the drilling contractors are consulted on this matter.

5.2.5 Batters

Table 4 presents recommended maximum batter angles for unsurcharged batters with heights less than 3.0m.

Table 4 Batter Angles

Material	Short Term	Long Term
Existing Fill	26°	18°
Controlled Fill	45°	26°
Clay Soils	45°	26°
Weathered Rock	60°	45°
Notes: 1. Subject to inspection by Soil Surveys Engineering. 2. Fill batter slopes are dependent on suitable compaction being achieved. 3. Batter angles assume no significant seepage.		

Inspection of cut batters should be carried out by a Geotechnical Engineer/Engineering Geologist from Soil Surveys Engineering, at the COMMENCEMENT OF BULK EXCAVATION WORKS, to confirm the stability of temporary cut batters.

It is essential that batters be suitably protected from erosion and scour by the establishment of ground cover and shrubs, installation of surface drains, etc. Runoff should not be allowed to discharge directly across the batters.

5.3 Site Classification

5.3.1 AS 2870 'Residential Slabs and Footings'

While a site classification in accordance with AS 2870 'Residential Slabs and Footings' relates to residential type construction and is not directly applicable for this development, it is, however, a valuable method of classification.

In assessing the site classification in accordance with AS 2870 'Residential Slabs and Footings', the following has been considered:-

- The subsurface profile (as per borehole records).
- Laboratory test results.
- Geotechnical data held by Soil Surveys Engineering.
- That the site is considered to be a reactive site (as per AS 2870-2011).
- That the site is subject to abnormal moisture conditions due to the presence of trees (on the building site and adjacent sites) and or proposed removal of trees (as per Clause 1.3.3 of AS 2870-2011).
- That the site is subject to abnormal moisture conditions due to the presence of an existing building and proposed removal of the existing building (as per Clause 1.3.3 of AS 2870-2011).
- Proposed earthworks, i.e. controlled fill greater than 0.4m is proposed.
- The presence of uncontrolled fill deeper than 0.4m.
- Likely future development.

Considering the above, and other factors, the site is classified Class 'P'.

5.3.2 Plumbing and Sanitary Drainage

It should be noted that the soil classification to AS 2870 for the purpose of design of plumbing and sanitary drainage systems (Form 1 Section 5) is 'H1'.

5.4 Site Reactivity

If this reactive site is assessed to be subject to ground surface movement predominantly due to soil reactivity **under normal moisture conditions**, a characteristic surface movement (y_s) within the Class 'H1' range ($y_s = \leq 55\text{mm}$) could be calculated (considering the proposed development, existing subsurface profile as per boreholes records, laboratory test results, and considering other relevant geotechnical data held by Soil Surveys Engineering); please note that tree induced movement would be additional to this value (refer below).

The foundation system for the proposed building should be designed and constructed to accommodate the potential ground movement resulting from the volume instability of the reactive clay soils.

Trees

Trees can greatly affect ground surface movement (greater movements than what would be expected may occur), and subsequently negatively impact structures.

The footing system should be designed for the effect of trees, with the Designer (Soil Surveys Engineering are not the Designer) considering the following cases (as appropriate):-

- Existing trees are left in place.

- Existing trees are removed prior to construction/development.
- New trees are subsequently planted.

AS 2870-2011 includes an informative section (refer Appendix H) on the design of footings for trees and in particular the calculations of the maximum potential surface movement due to the tree-induced suction change (that is in addition to the normal design suction profile). The Designer should refer to AS 2870-2011 for detailed discussion on this matter.

5.5 Building Foundations

5.5.1 Foundation Options

Given the subsurface profile encountered, the following foundation and slab on ground systems that could be considered include:-

- Foundation systems
 - high level footings
 - deep foundations (piers)
- Slab on ground floors
 - conventional industrial floor
 - stiffened raft
 - suspended slab

5.5.2 High Level Footings

High level strip/pad footings should found a minimum of 300mm into natural very stiff (or stronger) soils or weathered rock; footings will need to be deepened in areas of uncontrolled fill material (if any). Deepened footings, backhoe excavated pedestals (bulk piers) or bored piers may also be considered.

Table 5 outlines recommended allowable bearing capacities for high level footings.

Table 5 Allowable Bearing Capacities

Material		Allowable Bearing Capacity (kPa)	
		Strip Footing	Pad Footing
Fill		NR	NR
Clays	- Stiff to Very Stiff	200	250
	- Hard	350	400
Weathered Rock		400	400
Notes:-			
1. All capacities subject to inspection by Soil Surveys Engineering.			
2. NR = Not Recommended.			

5.5.3 Deep Foundations

Bored piers could be considered.

The design of a deep foundation system should consider the following:-

- Capacity
- Tensional effects due to swell
- Construction considerations

It is recommended that the deep foundation system on this project be designed in accordance with AS 2159-2009 'Piling - Design and Installation'. This code uses the limit state design method.

Compressional Capacity

The design of a single pile or a pile group must be such that both the geotechnical design strength $R_{d,g}$ and the structural design strength $R_{d,s}$ are greater than or equal to the design action effect E_d^* , i.e.

$$R_{d,g} \geq E_d \text{ and } R_{d,s} \geq E_d$$

The design geotechnical strength ($R_{d,g}$) can be calculated as the ultimate geotechnical design strength ($R_{d,ug}$) multiplied by the geotechnical strength reduction factor ϕ_g . Ultimate design geotechnical strength ($R_{d,ug}$) parameters for the materials encountered on the site are outlined in Table 6.

Table 6 Ultimate Geotechnical Strength (R_{ug}) Parameters

Material		fb - Base Bearing (kPa)	fm,s - Skin Friction (kPa)
Fill Material		NR	NC
Natural Clay	- Very Stiff	900	30
	- Hard	1800	60
Weathered Rock	- XW	1800	60
	- DW	2700	90

Notes:

1. NR = Not Recommended; NC = Not Considered in skin friction calculations.
2. Ultimate geotechnical strength for compression can be determined from $R_{d,ug} = fbAb + Asfm,s$.
3. Recommended geotechnical strength reduction factor (ϕ_g) - Refer Table 4.3.2(c) AS 2159.
4. Considering limit state analysis (AS 2159-2009), the design geotechnical strength R^*g is calculated by multiplying the ultimate design geotechnical strength $R_{d,ug}$ by the geotechnical strength reduction factor ϕ_g , i.e. $R_{d,g} = R_{d,ug} \times \phi_g$.
5. Should a "working stress" approach be adopted, a minimum factor of safety of 3.0 on base and 2.0 on skin friction is recommended.
6. Skin friction contribution from clays in the upper 1.5m of the profile should be ignored.
7. Piers should found at a minimum depth of four times the pier diameter and at least 2.00m below platform level.
8. The above parameters are for single piers. If piers are spaced at closer than three diameters, a reduction factor (Group Efficiency Ratio) may apply.
9. Base bearing and skin friction parameters to be confirmed by inspection by Soil Surveys Engineering.

Tensional Effects - Potential Uplift Due to Swelling Soils

In the case of structural design, the effect of swelling soils should be considered as part of the structural design of the foundation and as noted above. This can be calculated using the following formula (refer AS 2159-2009 3.3.2 (b)(i)(B)):-

$$S_u = 1.5F_{es}$$

Where S_u is the "Ultimate value of various actions appropriate for particular combinations" AS 2159-2009 Table 1.1 (refer also AS 1170) and F_{es} is the tensile (in this case) actions in the pile induced by vertical ground movements.

F_{es} can be calculated using the "Rational Pier Formula" as

$$F_{es} = 0.15 * SP * A_{mv}$$

- SP = Assessed Swell Pressure of soil (kPa)
- A_{mv} = Surface area of pile in intimate contact with swelling soils in the zone affected by swelling (i.e. zone of moisture variation = 1.50m)

Based on testing carried out on samples from the site, for design purposes $SP = 250\text{kPa}$.

Construction Considerations

Some difficulty with fall-in may occur with bored piers, particularly when drilling through fill material and clayey sand. It should be ensured that all loose material is removed from the base of piers prior to pouring of concrete. The use of a 'clean-out' bucket should be explicit in instructions to the drilling contractor. The practice of 'using water and spinning the augers' to remove loose material from the pier base is generally unacceptable.

Groundwater was not encountered in boreholes (refer Section 4.3) at the time of the investigation. However, should a bored pier foundation system be adopted, an allowance for dewatering and the use of liners should be made. **In addition, it may be prudent to drill a "trial pier" to fully assess construction difficulties.**

Some difficulty would be expected with screw piles being installed to levels below the zone of moisture variation due to soil strength. Conventional preboring is not recommended in reactive clay environments as the "prebored" zone may act as a conduit for moisture. Trial installations, reviewed by Soil Surveys Engineering, are recommended prior to acceptance of a screw pile system.

It should be noted that the use of screw piles in deep clay profiles may have issues with achieving typical installation torques (assuming this is the installation criteria adopted) particularly at the design foundation depths. It is therefore recommended that they be appropriately designed for the actual soil conditions using the parameters provided in this report. This may require an increased helix size or other structural design optimisation.

5.5.4 Slab on Ground

Field and laboratory test results indicate that the existing natural soils, along with any controlled fill, are suitable for slab on ground construction provided that earthworks are carried out in accordance with Section 5.2 and that appropriate engineering certification is provided.

However, the existing fill is considered to be not suitable in its current state to support the slab on ground floor. Should earthworks operations not result in a subgrade suitable for slab support, the slab on ground floor should be supported on pedestals/piers founding in competent natural soils.

Slab on ground floor options include:-

- Conventional industrial floor
- Fully suspended slab
- Stiffened raft

Industrial Floor

Provided that the performance criteria that would be normally applied to an industrial floor or pavement is consistent with the expected performance of a slab on ground floor for this project, an “industrial floor” could be adopted.

For design purposes, it is recommended that the slab on ground floor be designed adopting principles as contained in the Cement & Concrete Association of Australia, Industrial Floors & Pavements, Guidelines for Design, Construction & Specification.

For design of the industrial floor, a design CBR value of 2.0% is recommended. Inspection and testing should be carried out following stripping to fully assess subgrade conditions. A minimum subbase thickness of 300mm is recommended.

It is recommended that the slab on ground be suitably jointed and dowelled to allow some movement and also structurally separated from footings.

Fully Suspended

A fully suspended slab system involves the support of the slab on piers with provision of a void beneath the slab to accommodate the potential reactive clay movement; Figure 6 refers.

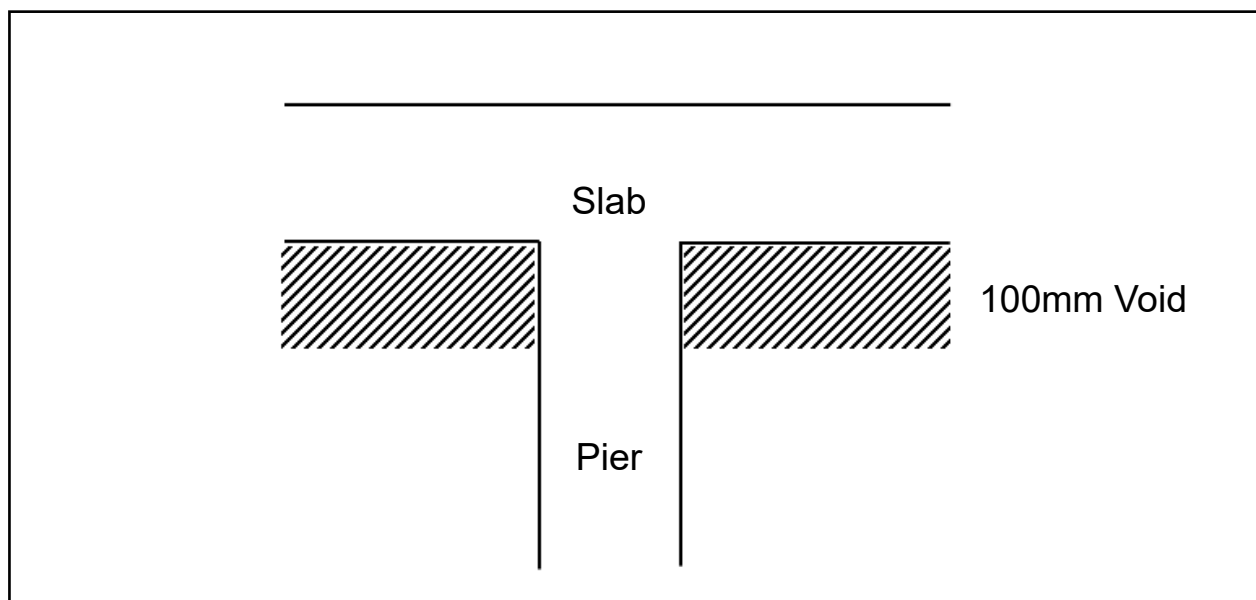


Figure 6

The design of piers should be in accordance with Section 5.5.3.

Stiffened Raft

A stiffened raft comprises perimeter and internal beams and is designed to accommodate potential ground surface movements.

This option does generally not suit large slab areas.

However, it may be prudent to consider use of a stiffened raft for movement sensitive slab areas (i.e. office, amenities, etc.) within the building.

5.5.5 Adjacent Feature/Excavation Considerations

Where proposed or existing (e.g. adjacent buildings) footings/piers are located adjacent to proposed or existing feature/excavations (e.g. retaining walls, underground service trenches, unsupported batters, etc.), the effect of the feature/excavation on proposed or existing footings/piers must be carefully considered. Deepening of proposed footings/piers is expected to be required in such circumstances; existing footings may need to be underpinned (or similar).

Soil Surveys Engineering and the Structural Engineer should be consulted on this matter prior to construction.

For underground service trenches (as a general guide) proposed footings and piers should extend to base a minimum of 500mm and 1.0m respectively below the feature/excavation base level for a distance of 1.0m out from the feature/excavation. Beyond 1.0m the footings and piers should be taken a minimum of 500mm and 1.0m respectively below an imaginary line (influence line) drawn up at 45° (1H:1V) from the feature/excavation base level. Note that actual depths (Soil Surveys Engineering should be consulted on this matter prior to construction) below the influence line will depend on footing/pier capacity and possible uplift considerations. Figure 7 refers.

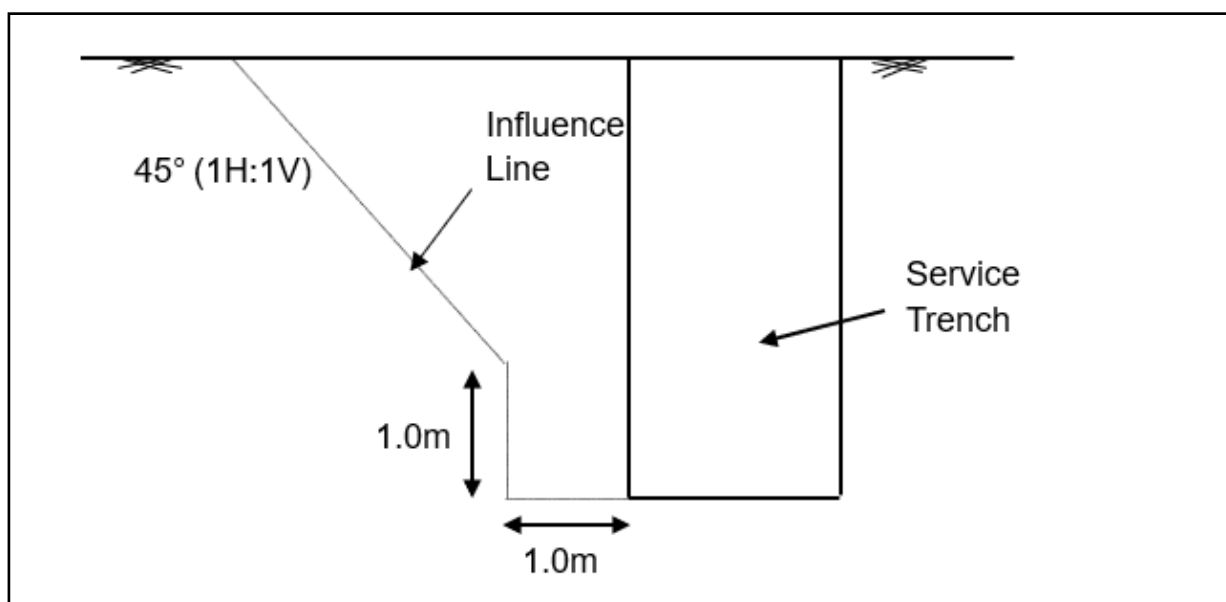


Figure 7

For all other feature types, Soil Surveys Engineering and the Structural Engineer should be consulted prior to construction.

For existing footings/piers that are located adjacent to a proposed feature/excavation, Soil Surveys Engineering and the Structural Engineer should also be consulted prior to construction.

5.5.6 Inspections

It is recommended that inspections be undertaken by an experienced and qualified geotechnical engineer following footing and during pier excavations to confirm the adequacy of the founding soils. Inspections should be carried out prior to placement of reinforcing steel and ordering of concrete.

5.6 Earthquake Design

With respect to structure design, the following would apply in accordance with AS 1170.4-2007:-

- Earthquake Hazard Factor - 0.08 (refer Figure 3.2 [F] of AS 1170.4-2007).
- Site Subsoil Class - Class 'Ce', Shallow Soil Site (refer Section 4.2 of AS 1170.4-2007).

5.7 Articulation and Detailing

Given the potentially reactive nature of the subsurface material, the importance of architectural and structural detailing cannot be over-emphasised. It is essential that the building be suitably articulated/detailed to allow for ground surface movement. In addition, the overall design of the building and surrounds should consider the reactive soils, with the objective being to maintain a stable moisture environment. Your attention is drawn to recommendations contained in Section 5.8.

It is recommended that any masonry walls be articulated. This articulation may be achieved by the use of full height (footings to eaves) openings or vertical construction joints at regular intervals.

Guidelines on articulation are contained in the Cement and Concrete Association's Technical Note 61, 'Articulated Walling'.

5.8 Site Management

It is important that proper site management methods be observed for the existing soil conditions by both the builder at the time of the construction and the owner(s) throughout the life of the proposed development.

Particular reference to site management matters is set out in AS 2870-2011. It should be noted that where proper site management, particularly with respect to change in moisture conditions, is not followed, the foundation recommendations contained in this report could be considered void.

The following are some specific comments with respect to site management and the reader is also directed to CSIRO publication, "Foundation Maintenance and Footing Performance: A Homeowners Guide" (Building Technology File 18) for further information:-

- It is important that the site be well drained. The ground around the structures should slope away at 1 in 50 for 3 metres and then fall to the stormwater system to prevent ponding of water adjacent to the structures.
- Roof downpipes and garden taps should not be allowed to saturate founding soils.
- It is recommended that service trenches under the building be minimised, as disruption or breakage of these service lines could lead to saturation of the under slab material resulting in an increase to moisture condition. If this were to occur, all footing recommendations contained within this report would be void.
- Footings should be placed with minimal delay after excavation to avoid desiccation or wetting of the founding soils. If footings cannot be poured on the same day as excavation, a blinding layer of 75mm thickness is recommended. Piers should be poured immediately.
- Do not let the slab subgrade "dry out" prior to casting.
- Future shrubs and trees should be planted at a distance at least equivalent to their mature height away from the building to avoid shrinkage movement in the potentially expansive founding soils. Existing trees that may encroach this restriction should be removed as soon as possible prior to commencement of construction. Replenishment of soil moisture to equilibrium conditions after tree removal can result in soil heave. Also refer Section 5.3.

5.9 Retaining Walls

It is recommended that any retaining structure be designed in accordance with AS 4678-2002 'Earth Retaining Structures'. Section 3 of this code outlines the design requirements for retaining structures which specifically includes both Ultimate and Serviceability Limit Modes. It is recommended that the retaining structures be assessed for each of these modes.

Please note that the following section provides general recommendations with respect to some of these limit modes, however, it doesn't provide a detailed assessment in full accordance with the above mentioned modes for AS 4678. A detailed assessment can be undertaken of the geotechnical

aspects of the Limit Modes for the proposed retaining structures once details of the wall have been provided.

For cantilever walls, or temporarily braced walls, which allow movement at the top, i.e. at least 0.005H in clays, the active case (K_a) applies with a triangular distribution. For walls which cannot tolerate this movement, the at rest (K_o) case applies with a triangular distribution.

Lateral pressure assessment must also consider hydrostatic pressure and surcharge loadings.

The following parameters (unfactored) may be adopted for wall design (Table 7).

Table 7 Parameters for use in Retaining Wall Design

Material	Density (kN/m ³)	Earth Pressure Coefficient Vertical Wall			Long Term Drained ϕ
		K_a	K_o	K_p	
Existing Fill	17	0.41	0.58	NR	25°
Controlled Fill	19	0.36	0.53	2.70	28°
Clay	19	0.36	0.53	2.70	28°
Weathered Rock	20	0.31	0.47	3.20	32°
Notes: 1. NR - Not Recommended.					

Any backfill placed behind the wall should be loose granular material. The backfill should not be heavily compacted since research has shown that compaction can raise the earth pressure to above the 'at rest' pressure.

Adequate surface and subsoil drainage should be provided for all retaining walls on the site. Cut-off / interceptor drains should also be provided around the high side of the wall to ensure stormwater runoff from the area above the wall is suitably diverted.

The placement of a filter fabric between the retained soil and the drainage material (e.g. granular backfill) for protection against silting of the drainage material is recommended. The outlets to subsoil pipe drains must be located beyond the ends of the walls and connected to a proper drainage system. It is suggested the pipes be wrapped in filter fabric to minimise silting.

In weather exposed locations, to reduce infiltration by surface runoff, the surface of the backfill should be sealed. This can be achieved by either compacting a material of low permeability, i.e. on site clay, or concrete, with a slope towards an open drain.

Due to possible long term problems with blocking of gravel filters and drains and short term storm conditions that could flood the fill behind retaining walls, it is recommended that all retaining walls be designed for some water pressure distribution.

During installation of any retaining walls, the insitu soils should be battered back to minimise fall-in and subsequent disruption of works. Suitable precautions to satisfy Health & Safety requirements must also be adhered to.

5.10 Pavements

5.10.1 General

It is understood that a flexible pavement is proposed for the majority of the driveway and carparking areas along with some rigid pavement areas.

Following earthworks, it is envisaged that a Silty Clay subgrade will be exposed.

Laboratory testing was carried out on the natural clay material sampled from Boreholes BH04, BH06 and BH08 to determine this material's California Bearing Ratio (CBR); the CBR value represents the 'strength' of this material, compacted to 100% standard density ratio, under saturation. The Soaked CBR test produced values of between 2.0% and 3.5% with % Swells of 2.5% to 4.5%.

Given the above and considering the reactive nature of the subgrade soils, a maximum design CBR value of 2.0% and a modulus of subgrade reaction of 20kPa/mm - Austroads (2009)¹ are recommended for design.

5.10.2 Design for Reactive Soils ($H \leq 2.3\text{m}$)

Pavement design should also consider the presence of reactive soils.

Testing indicates subgrade swell values 2.5% to 4.5%.

A review of the DTMR document, Supplement to 'Part 2: Pavement Structural Design' of the Austroads Guide to Pavement technology suggest the following approach:-

- Assess the expansive nature of the subgrade (Table 5.3.5 in DTMR document):-
 - CBR Swell <0.5% - Low
 - CBR Swell 0.5% to 2.5% - Moderate
 - CBR Swell 2.5% to 5.0% - High
 - CBR Swell 5.0% to 10.0% - Very High
 - CBR Swell >10.0% - Extreme
- The DTMR document noted that in the case of Low or Moderate the document indicates that additional cover beyond that provided by the pavement and support layers is not typically required.
- The DTMR recommendation for highly and very highly expansive soil, the use of Table 5.3.5 (reproduced below) is recommended.

¹ Austroad (2009) Review of relationship to predict Subgrade Modulus from CBR, Pub No. AP-T130/09.

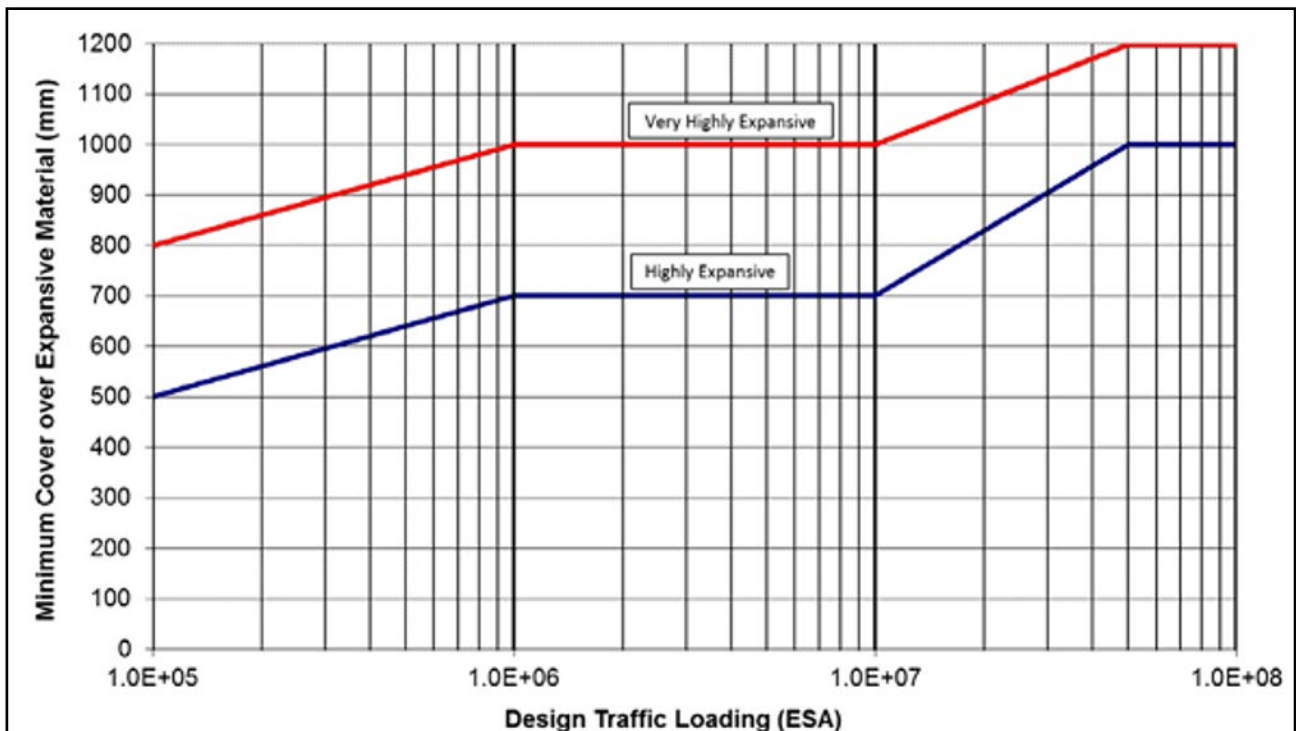


Figure 8 - Typical cover thickness over highly and very expansive material for flexible pavements (thickness includes the pavement)

5.10.3 General Recommendations

Along with recommendations contained in Sections 5.1 and 5.2, the following general earthworks recommendations are made:-

- i) Incorporate a perimeter drain at the pavement edges to prevent possible deterioration under wet weather.
- ii) Pavements should be well drained both during and upon completion of construction. Water should not be allowed to pond on or near pavement surfaces.
- iii) Pavement gravel should comply with the local council quality specifications for sub base and base course material.
- iv) Subgrades should be compacted to achieve the minimum density ratios as outlined in Section 5.2 'Earthworks'.
- v) It is recommended that inspection and testing be carried out following general earthworks to fully assess subgrade conditions.

6.0 Limitations

We have prepared this report for the use of **Thomas Adsett**, for design purposes in accordance with generally accepted geotechnical engineering practices. No other warranty, expressed or implied, is made as to the professional advice included in this report. This report has not been prepared for use by parties other than **Thomas Adsett**; it may not contain sufficient information for purposes of other parties or for other uses. Please note that any third party relying on the information contained in this report for any purpose whatsoever does so entirely at its own risk, and any duty of care to that third party is excluded.

Any interpretation or recommendation given by Soil Surveys Engineering shall be understood to be based on judgement and experience and not on greater knowledge of the facts than the reported investigations would imply. The interpretation and recommendations are therefore opinions provided for our client's sole use in accordance with the specific brief. As such they do not necessarily address all aspects of ground behaviour on the subject site. Information provided by others has been taken in good faith, but no liability can be accepted for information provided by others.

Your attention is drawn to 'Appendix A', 'Notes Relating to this Report'. Interpretation of factual data given in this report is based on judgement, not a greater knowledge of facts other than those reported.

Interpretation of the information shown on the logs, and its application to design and construction, should therefore consider the spacing and depth of test locations, the method of investigation, the frequency of sampling and testing and the possibility of other than "straight line" variations between the test locations. Subsurface conditions between and below test locations may vary significantly from conditions encountered at the test locations.

Please note that should, following detailed design, footings or piers/piles be required to extend within 3B (B = footing width/pier diameter) above the termination level of the boreholes/DCP or any excavations extend below borehole/DCP termination levels, Soil Surveys Engineering should be contacted immediately. In this case, geotechnical data in this report should be considered preliminary only; additional investigation is likely to be required.

If conditions encountered on site during construction appear to vary from those expected from the information contained in the report, the Company strongly recommends that it immediately be notified. Most problems are more readily resolved when conditions are exposed than at some later stage, after the event. Should Soil Surveys Engineering not be notified or if this notification is delayed, then Soil Surveys cannot be held responsible for the affect that any variation has on any aspect of the development.

Soil Surveys Engineering consider that a documentation review service (during the design phase and prior to construction) to verify that the intent of geotechnical recommendations is properly reflected in the design, along with construction inspections, forms a very important component of the geotechnical engineering design service/process.

The geotechnical review ensures geotechnical risks to our client and their project are minimised at the design and tender stage of the project. Further, with Soil Surveys Engineering being commissioned to carry out geotechnical construction inspections, an opportunity at the time of

construction to confirm any assumptions made in the preparation of the report and allow the effect of any normally occurring variation in ground conditions to be assessed with respect to construction becomes available.

The above statements are not intended to reduce the level of responsibility accepted by Soil Surveys Engineering in accordance with our commission, but rather to ensure that all parties who may rely on this report are aware of the responsibilities each assumes in doing so and the risks they accept should they decline to have Soil Surveys Engineering carry out a geotechnical documentation review and geotechnical construction inspections.

It is highly recommended that the Client avail themselves of these review and inspection services; our standard rates will apply.



Cameron Johnson (RPEQ 7052)

Principal Geotechnical Engineer

For and on behalf of

Soil Surveys Engineering Pty Limited

Appendices



Appendix A

Notes Relating to this Report



NOTES RELATING TO THIS REPORT

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INTRODUCTION

These notes are provided by Soil Surveys Engineering Pty Limited (the Company) to complement the geotechnical report in regard to classification methods and field procedures. Not all notes are necessarily relevant to all reports.

The ground is a product of continuing natural and man-made processes and therefore exhibits a variety of characteristics and properties which vary from place to place and can change with time. Geotechnical engineering involves gathering and assimilating limited information about these characteristics and properties in order to understand or predict the behaviour of the ground on a particular site under certain conditions. This report may contain such information obtained by inspection, excavation, probing, sampling, testing or other means of investigation. If so, they are directly relevant only to the ground at the place where and at the time when the investigation was carried out.

DESCRIPTION AND CLASSIFICATION METHODS

Soils - The methods of description and classification of soils and rocks used in this report are based on Australian Standard 1726-2017 (Geotechnical Site Investigations), where appropriate. In general, descriptions cover the following properties - soil or rock type, colour, structure, strength or density, and inclusions. Identification and classification of soil and rock involves judgement and the Company infers accuracy only to the extent that is common in current geotechnical practice.

Soil types are described according to the dominant particle size and behaviour as set out in AS 1726-2017.

Cohesive soils are classified on the basis of strength (consistency) either by use of hand penetrometer, shear vane, laboratory testing or engineering examination. The strength terms are defined in AS 1726-2017 Table 11.

Non-cohesive soils are classified on the basis of relative density usually based on insitu testing or engineering examination (see AS 1726-2017 Table 12).

Rocks - Rock types are classified by their geological names (AS 1726-2017 Tables 15 to 18), together with descriptive terms regarding weathering (AS 1726-2017 Table 20), strength (AS 1726-2017 Table 19), defects (AS 1726-2017 Table 22), etc.

SAMPLING

Sampling is carried out during drilling or from other excavations to allow engineering examination (and laboratory testing where required) of the soil or rock.

Disturbed samples taken during drilling provide information on plasticity, grain size, colour, moisture content, minor constituents and, depending upon sample disturbance, (information on strength and structure).

Undisturbed samples are taken by pushing a thin walled sample tube, usually 50mm diameter (U50), into the soil and withdrawing it with a sample of the soil contained in a relatively undisturbed state. Such samples yield information on structure and strength, and are necessary for laboratory

determination of shear strength, volume change potential and compressibility. Undisturbed sampling is generally effective only in cohesive soils.

Details of the type and method of sampling used are given on the attached logs.

SAMPLE STORAGE – SOIL, ROCK AND WATER

SAMPLES

Soil samples (not subject to testing) are not stored beyond a period of 90 days of taking or receiving said soil sample. Rock core (not subject to testing) is not stored beyond a period of six months of taking or receiving said rock core.

Should any party require that soil samples (not subject to testing) be stored beyond 90 days, or rock core (not subject to testing) be stored beyond six months, please contact Soil Surveys Engineering.

Water samples (not subject to testing) are not stored beyond a period of seven days of taking or receiving water samples.

TEST LOCATIONS

Test locations (e.g. boreholes, CPT's, test pits etc.) were based on available access at the time of testing. Test locations may have been shifted if access was not suitable.

Unless noted otherwise, accuracy of test locations are to the accuracy of hand held GPS equipment.

INVESTIGATION METHODS

The following is a brief summary of investigation methods currently adopted by the Company and some comments on their use and application.

Test Pits - These are normally excavated with a backhoe or a tracked excavator, allowing close examination of the insitu soils if it is safe to descend into the pit. The depth of penetration is limited to approximately 3.0m for a backhoe and up to 6.0m for an excavator. Limitations of test pits are the problems associated with disturbance and difficulty of reinstatement and the consequent effects on close-by structures. Care must be taken if construction is to be carried out near test pit locations to either properly recompact the backfill during construction or to design and construct the structure so as not to be adversely affected by poorly compacted backfill at the test pit location.

Hand Auger Drilling - A borehole of 50mm to 100mm diameter is advanced by manually operated equipment. Refusal of the augers can occur on a variety of materials such as hard clay, gravel or rock fragments and does not necessarily indicate rock level.

Continuous Spiral Flight Augers - The borehole is advanced using 75mm to 300 mm diameter continuous spiral flight augers, which are withdrawn at intervals to allow sampling or insitu testing. This is a relatively economical means of drilling in clays and in sands above the water table. Samples are returned to the surface by the flights or may be collected after withdrawal of the augers. Information from the drilling (as distinct from specific sampling) is of relatively lower reliability due to remoulding, inclusion of cuttings from above or softening of samples by groundwater, or

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uncertainties as to the original depth of the samples. Augering below the groundwater table has a lower reliability than augering above the water table. Various drill bits are attached to the base of the augers during the drilling. The depth of refusal of the different bit types can provide information as to the strength of the material encountered. Generally the 'TC' bit (a tungsten carbide tipped screw type bit) is used.

Wash Boring - The borehole is usually advanced by a rotary bit with water or fluid pumped down the hollow drill rods and returned up in the space between the rods and the soil or casing, carrying the drill cuttings. Only major changes in stratification can be determined from the cuttings, together with some information from "feel" and rate of penetration. More accurate information on soil strata is gained by regular testing and sampling using the Standard Penetration Test (SPT) and undisturbed thin walled tube samples (U50).

Mud Stabilized Drilling - Either Wash Boring or Continuous Core Drilling can use drilling mud as a circulating fluid to stabilize the borehole. The term "mud" encompasses a range of products ranging from bentonite to polymers such as Revert or Biogel. The mud tends to mask the cuttings and reliable identification is only possible from regular intact sampling (e.g. from SPT and U50 samples) or from rock coring, etc.

Continuous Core Drilling - A continuous core sample is obtained using a diamond or tungsten carbide tipped core barrel. Provided full core recovery is achieved (which is not always possible in very weak rocks and granular soils), this technique provides a very reliable method of investigation. In rocks, NMLC coring (nominal 52 mm diameter) is usually used with water flush. The length of core recovered is compared to the length drilled and any length not recovered is shown as CORE LOSS. The location of losses is determined on site by the supervisor. If the location of the loss is uncertain, it is placed at the top end of the run, when the core is placed in a storage tray and recorded on the log.

Standard Penetration Tests - Standard Penetration Tests (SPT) are used mainly in non-cohesive soils, but can also be used in cohesive soils, as a means of indicating density or strength. The test procedure is described in Australian Standard 1289, "Methods of Testing Soils for Engineering Purposes" - Test 6.3.1.

The test is carried out in a borehole by driving a 50mm diameter split sample tube with a tapered shoe, under the impact of a 63 kg hammer with a free fall of 760 mm. It is normal for the tube to be driven in three successive 150 mm increments and the 'N' value is taken as the number of blows for the last 300 mm, the upper 150 mm being neglected due to possible disturbance from the drilling method. In dense sands, very hard clays or weak rock, the full 450 mm penetration may not be practicable and the test is discontinued at a reduced penetration.

In the case where full penetration is obtained with successive blow counts for each 150 mm of, say 4, 6 and 7 blows, the record shows,

4, 6, 7

N = 13

In a case where the test is discontinued short of full penetration, say after 15 blows for the first 150 mm and 30 blows for the next 40 mm, the record shows:

15, 30/40mm

The results of the test can be related empirically to the engineering properties of the soil.

Occasionally, the drop hammer is used to drive 50mm diameter thin walled sample tubes (U50) in clays. In such circumstances, it is noted on the borehole logs.

A modification to the SPT test is where the same driving system is used with a solid 600 tipped steel cone of the same diameter as the SPT hollow sampler. The solid cone can be continuously driven for some distance in soft clays or loose sands, or may be used where damage would otherwise occur to the SPT. The results of this Solid SPT are shown as "N_c" on the borehole logs, together with the number of blows per 150 mm penetration.

Cone Penetration Tests - Test Method - Cone Penetration Tests (CPT) are carried out in accordance with AS 1289 Test 6.5.1-1999, using an electrical friction-cone penetrometer.

The test essentially comprises the measurement of resistance to penetration of a cone of 35.7 mm diameter pushed into the soil at a rate of 10-20 mm per second by hydraulic force. The resistance to penetration is recorded in terms of pressure on the end area of the cone (cone resistance, q_c , in MPa) and friction on the side of the 135 mm long sleeve immediately above the top of the cone (friction resistance, f_s , in kPa). These forces are measured by electrical transducers (strain gauges) within the cone device. The ratio between friction resistance and cone resistance is also calculated as a percentage, i.e.-

$$\text{Friction Ratio (FR)} = \frac{\text{Friction Resistance, } f_s \text{ (kPa)} \times 100}{\text{cone resistance, } q_c \text{ (kPa)}}$$

The friction ratio, FR, is generally low in sands (less than 1% or 2%) and generally higher in clays (say 3% or more). The interpretation of sandy clays, clayey sands and material with a high silt content is more difficult, but intermediate values (between 1% and 3%) would be expected. Highly organic clays and peats generally have a friction ratio in excess of 5%.

Static cone data is recorded in the field on disc for later presentation using computer aided drafting.

The equipment can be operated from any conventional drill rig. A total applied load in the range of 4 to 10 tonnes is required for practical purposes, although lighter loads may be used. The cone penetrometers are available with various capacities of cone resistance ranging up to 100 MPa for general purpose investigations, while a range of 0 to 10 MPa can be used where more sensitive investigations of soft clay are required.

The cone resistance value provides a continuous measure of soil strength or density, and together with the friction ratio, provide very useful indications of the presence of narrow bands of geotechnically significant layers such as thin, soft clay layers or lenses of sand which might otherwise be missed using conventional drilling methods.

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The lithology of the encountered soils is interpreted from static cone data and is generally presented on the static cone log sheets.

It is important to note that the lithology is interpreted information and is based on research by Schmertmann (1970), Sanglerat (1972), Robinson and Campinalli (1986), modified to suit local conditions as indicated by borehole information and laboratory testing.

As soils generally change gradually it is sometimes difficult to accurately describe depths of strata changes, although greater accuracy is obtained with the static cone compared with conventional drilling. In addition, friction ratios decrease in accuracy with low cone resistance values, and in desiccated soils. As a result, some overlap and minor discrepancies may exist between static cone and nearby borehole information.

Portable Dynamic Cone Penetrometers - Portable Dynamic Cone Penetrometer (DCP) tests are carried out by driving a rod into the ground with a falling weight hammer and measuring the blows for successive 100mm increments of penetration.

The DCP comprises a Cone of 20 mm diameter with 30 degree taper attached to steel rods of smaller section.

The cone end is driven with a 9 kg hammer falling 510 mm (AS 1289 Test 6.3.2). The test was developed initially for pavement subgrade investigations, and empirical correlations of the test results with California Bearing Ratio have been published by various Road Authorities. The Company has developed their own correlations with Standard Penetration tests and Density Index tests in sands.

LOGS

The borehole or test pit logs presented herein are an engineering and/or geological interpretation of the subsurface conditions, and their reliability will depend to some extent on the frequency of sampling and the method of drilling or excavation. Ideally, continuous undisturbed sampling or core drilling will enable the most reliable assessment but is not always practicable or possible to justify on economic grounds. In any case, the boreholes or test pits represent only a very small sample of the total subsurface conditions.

The attached explanatory notes define the terms and symbols used in preparation of the logs.

Interpretation of the information shown on the logs, and its application to design and construction, should therefore take into account the spacing of boreholes or test pits, the method of drilling or excavation, the frequency of sampling and testing and the possibility of other than "straight line" variations between the boreholes or test pits. Subsurface conditions between boreholes or test pits may vary significantly from conditions encountered at the borehole or test pit locations.

GROUNDWATER

Where groundwater levels are measured in boreholes, there are several potential problems.

- Although groundwater may be present in lower permeability soils, it may enter the hole slowly or perhaps not at all during the time the hole is open.
- A localized perched water table may lead to an erroneous indication of the true water table.
- Water table levels will vary from time to time with seasons or recent weather changes and may not be the same at the time of construction.
- The use of water or mud as a drilling fluid will mask any groundwater inflow. Water has to be bailed out of the bore and mud must be washed out of the hole or "reverted" if water observations are to be made.

More reliable measurements can be made by use of standpipes which are read after stabilizing at periods ranging from several days to perhaps weeks for low permeability soils. Piezometers, sealed in a particular stratum, may be advisable in low permeability soils or where there may be interference from perched water tables or surface water.

FILL

The presence of fill materials can often be determined only by the inclusion of foreign objects (e.g. bricks, steel, etc.) or by distinctly unusual colour, texture or fabric. Identification of the extent of fill materials will also depend on investigation methods and frequency. Where natural soils similar to those at the site are used for fill, it may be difficult with limited testing and sampling to reliably determine the extent of the fill.

The presence of fill materials is usually regarded with caution as the possible variation in density, strength and material type is much greater than with natural soil deposits. Consequently, there is an increased risk of adverse engineering characteristics or behaviour. If the volume and quality of fill is important to a project, then frequent test pit excavations are preferable to boreholes.

LABORATORY TESTING

Laboratory testing is normally carried out in accordance with Australian Standard 1289 "Methods of Testing Soil for Engineering Purposes". Details of the test procedure used are given on the individual report forms and the attached explanatory notes summarize important aspects of the Laboratory Test Procedures adopted.

ENGINEERING REPORTS

Engineering reports are prepared by qualified personnel and are based on the information obtained and on current engineering standards of interpretation and analysis. The information provided in Soil Surveys Engineering reports is opinion and interpretation and not factual. The client/contractor increases their risk by not retaining the person who authored the geotechnical report, to carry out site inspection and review (overseeing role) during construction, to confirm opinion and interpretation expressed in the report is accurate. Where the report has been prepared for a specific design proposal the information and interpretation may not be relevant if the design proposal is changed. If this happens, the Company will be pleased to

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review the report and the sufficiency of the investigation work.

Every care is taken with the report as it relates to interpretation of subsurface conditions, discussion of geotechnical aspects and recommendations or suggestions for design and construction. Since the test sites in any exploration represent a very small proportion of the total site and since the exploration only identifies actual ground conditions at the test sites, even under the best circumstances actual conditions may vary from those inferred to exist. No responsibility is taken for:-

- Unexpected variations in ground and/or groundwater conditions.
- Changes in policy or interpretation of policy by statutory authorities.
- The actions of other persons.
- Any work where the company is not given the opportunity to supervise the construction using the Companies designs/recommendations.

If differences occur, the Company will be pleased to assist with investigation or advice to resolve any problems occurring.

SITE ANOMALIES

In the event that conditions encountered on site during construction appear to vary from those expected from the information contained in the report, the Company requests that it immediately be notified. Most problems are more readily resolved when conditions are exposed than at some later stage, well after the event.

Extreme events including but not limited to the results of climate change, e.g. flood levels above previously identified levels, beach scour or erosion beyond normal expectations (as identified by local authorities) extreme rainfall events, war, espionage, sabotage may result in different conditions between time of investigation and time of construction.

REPRODUCTION OF INFORMATION FOR

CONTRACTUAL PURPOSES

Attention is drawn to the document "Guidelines for the Provision of Geotechnical Information in Construction Contracts (1987)", published by the Institution of Engineers, Australia. Where information obtained from this investigation is provided for tendering purposes, it is recommended that all information, including the written report and discussion, be made available. In circumstances, where the discussion or comments section is not relevant to the contractual situation, it may be appropriate to prepare a specially edited document. The Company would be pleased to assist in this regard and/or to make additional report copies available for contract purposes at a nominal charge.

REVIEW OF DESIGN

Where major civil or structural developments are propose or where only a limited investigation has been completed or where the geotechnical conditions/ constraints are quite

complex, it is prudent to have a joint design review which involves a senior geotechnical engineer. We would be happy to assist in this regard as an extension of our investigation commission. Construction drawings should be reviewed by Soil Surveys Engineering, with sufficient time to allow changes if required, prior to inspections. Otherwise Soil Surveys Engineering reserves the right to refuse to carry out inspections.

SITE INSPECTION

The Company will always be pleased to provide engineering inspection services for geotechnical aspects of work to which this report is related.

- i. Site visits during construction to confirm reported ground conditions
- ii. Site visits to assist the contractor or other site personnel in identifying various soil/rock types such as appropriate footing or pier founding depths, the stability of a filled or excavated slope; or
- iii. Full-time engineering presence on site.

In the vast majority of cases it is advantageous to the principal for the geotechnical engineer who wrote the investigation report to be involved in the construction stage of the project.

The geotechnical engineer cannot take responsibility for variations in encountered conditions, where he is not given the opportunity to review plans for the proposed development with sufficient time to allow review and make changes to the proposed development if required, and where he is not given the opportunity to inspect the site and oversee construction methods with regard to site conditions with sufficient time to observe all relevant site conditions and operations.

RESPONSIBLE USE OF GEOTECHNICAL INFORMATION

Recommendations in our report are for design purposes only and provided on the basis that inspections are carried out to allow finalisation of opinions and recommendations contained in our report.

The geotechnical investigation consisting of field and laboratory testing has been carried out to indicate typical conditions by indicating conditions and parameters at the specific locations of boreholes/test pits. Subsurface conditions are indicated at these locations only and the inference of conditions between or away from these locations (interpolation and extrapolation) involves a certain degree of risk. Persons inferring such conditions or carrying out such inferences should do so with a degree of caution and conservatism which is commensurate with the consequences of the risk of error.

Estimates of volumes based on our findings require interpolation and extrapolation between test locations and as such may be significantly different from actual volumes.

Appendix B

Borehole Record Sheets



Job No : 1-29688
 Client : Thomas Adsett
 Project : CQU ROK Tafe Consolidation
 Location : Central Queensland University, Norman Gardens.

Easting : 246276.00
 Northing : 7418630.00
 RL : N/A
 Drill Rig : CDR-2

Sheet : 1 OF 1
 Logged : Rick Hayward
 Logged Date : 09/12/2025
 Checked : CJ

Drilling Method	Water	Depth (m)	Graphic Log	Material Description	DCP	Samples and Remarks
Hand Auger		0.08		FILL Clayey SAND (SC): loose, fine to coarse grained, dark brown, medium to high plasticity clay, trace fine to medium gravel, moist, with organics.		
		0.5		NATURAL Silty CLAY (CH): very stiff, high plasticity, dark brown, trace fine to coarse sand, trace fine gravel, moist.		0.40-0.70, D
		1.0		NATURAL Sandy CLAY (CL-CI): low to medium plasticity, light brown, fine to coarse grained sand, with fine to medium gravel, moist.		0.60-0.90, U50 PP280
		1.5		As above: hard, tending to sandstone xw.		1.30-1.50, D
		2.0				
		2.5		Extremely weathered SANDSTONE (XW): recovered as Clayey Sand - very dense, fine to coarse grained, dark brown, low to medium, moist.		
		3.0				
		3.5		NATURAL Sandy CLAY (CL-CI): hard, low to medium plasticity, dark brown, fine to coarse grained sand, with fine to medium gravel, moist.		
		4.0				4.00-4.50, D
				BH01 Terminated at 4.5 m		

Comments

1. Groundwater not encountered

Key

U - Undisturbed Tube
 D - Disturbed (small)
 B - Bulk Disturbed (large)
 ASS - Acid Sulfate Soil
 PP - Pocket Penetrometer
 VS - Vane Shear
 XW - Extremely Weathered
 DW - Distinctly Weathered
 ☒ Groundwater Observed
 ☒ Standing Water

Approved By: CJ

Approval Date: 14/01/2026

Job No : 1-29688
 Client : Thomas Adsett
 Project : CQU ROK Tafe Consolidation
 Location : Central Queensland University, Norman Gardens.

Easting : 246346.00
 Northing : 7418641.00
 RL : N/A
 Drill Rig : CDR-2

Sheet : 1 OF 1
 Logged : Rick Hayward
 Logged Date : 10/12/2025
 Checked : CJ

Drilling Method	Water	Depth (m)	Graphic Log	Material Description	DCP	Samples and Remarks
Auger Drilling		0.025		PAVEMENT Asphalt.		
		0.25		FILL Sandy Silty GRAVEL (GM): medium dense, fine to medium sized, dark brown, fine to coarse grained sand, moist.		0.25-0.50, D
		0.5		FILL Sandy CLAY (CI-CH): very stiff, medium to high plasticity, dark brown, fine to coarse grained sand, with fine to medium gravel, moist.		
		0.7		NATURAL Silty CLAY (CH): very stiff, high plasticity, dark grey, trace fine to medium sand, trace fine gravel, moist, trace of organics.		1.00-1.30, U50 PP260
		1.0				
		1.5				
		2.0				
		2.3		NATURAL Sandy CLAY (CL-CI): hard, low to medium plasticity, dark brown, fine to coarse grained sand, with fine to medium gravel, moist.		2.50-2.80, D
		2.5				
		2.8		NATURAL Clayey Sandy GRAVEL (GC): dense to very dense, fine to medium sized, dark brown, fine to coarse grained sand, low to medium plasticity clay, moist.		
	3.0		NATURAL Clayey Sandy GRAVEL (GC): dense to very dense, fine to medium sized, dark brown, fine to coarse grained sand, low to medium plasticity clay, moist.			
	3.3		NATURAL Gravelly CLAY (CI): hard, medium plasticity, dark brown, fine to medium sized gravel, with fine to coarse sand, moist.			
	3.5					
	4.0		Extremely weathered SILTSTONE (XW): recovered as Clayey Gravel - very dense, fine to medium grained, dark brown, low, moist.			
			BH02 Terminated at 4.5 m			

Comments

1. Groundwater not encountered

Key

U - Undisturbed Tube
 D - Disturbed (small)
 B - Bulk Disturbed (large)
 ASS - Acid Sulfate Soil
 PP - Pocket Penetrometer
 VS - Vane Shear
 XW - Extremely Weathered
 DW - Distinctly Weathered
 Groundwater Observed
 Standing Water

Approved By: CJ

Approval Date: 14/01/2026

Job No : 1-29688
 Client : Thomas Adsett
 Project : CQU ROK Tafe Consolidation
 Location : Central Queensland University, Norman Gardens.

Easting : 246291.00
 Northing : 7418569.00
 RL : N/A
 Drill Rig : CDR-2

Sheet : 1 OF 1
 Logged : Rick Hayward
 Logged Date : 10/12/2025
 Checked : CJ

Drilling Method	Water	Depth (m)	Graphic Log	Material Description	DCP	Samples and Remarks
Auger Drilling		0.03			0 5 10 15 20 25	
		0.2		PAVEMENT Asphalt.		
		0.5		FILL Sandy Silty GRAVEL (GM): medium dense, fine to coarse sized, dark brown, fine to coarse grained sand, moist.		0.30-0.70, D
		0.8		NATURAL Silty CLAY (CH): very stiff, high plasticity, dark grey, trace fine to coarse sand, trace fine gravel, moist.		0.60-0.90, U50 PP>600
		1.0		NATURAL Gravelly CLAY (CL-CI): hard, low to medium plasticity, light brown, fine to medium sized gravel, with fine to coarse sand, moist.		
		1.1		NATURAL Clayey GRAVEL (GC): dense, fine to medium sized, dark brown, low to medium plasticity clay, with fine to coarse sand, moist.		1.30-1.50, D
		1.5		Extremely weathered SILTSTONE (XW): recovered as Clayey Gravel - very dense, fine to medium grained, light brown, low, moist.		
		1.6				
		2.0				
		2.4		SILTSTONE (DW): distinctly weathered, low strength, light brown mottled light grey, fine to medium grained, moist.		
			BH03 Refusal at 2.5 m			

Comments

- Groundwater not encountered
- Borehole terminated at 'TC' bit refusal

Key

U - Undisturbed Tube
 D - Disturbed (small)
 B - Bulk Disturbed (large)
 ASS - Acid Sulfate Soil
 PP - Pocket Penetrometer
 VS - Vane Shear
 XW - Extremely Weathered
 DW - Distinctly Weathered
 Groundwater Observed
 Standing Water

Approved By: CJ

Approval Date: 14/01/2026

Job No : 1-29688
 Client : Thomas Adsett
 Project : CQU ROK Tafe Consolidation
 Location : Central Queensland University, Norman Gardens.

Easting : 246305.00
 Northing : 7418614.00
 RL : N/A
 Drill Rig : CDR-2

Sheet : 1 OF 1
 Logged : Rick Hayward
 Logged Date : 10/12/2025
 Checked : CJ

Drilling Method	Water	Depth (m)	Graphic Log	Material Description	DCP	Samples and Remarks
Hand Auger		0.18		FILL Sandy Silty GRAVEL (GM): medium dense, fine to medium sized, dark brown, fine to coarse grained sand, moist.		
		0.5		NATURAL Silty CLAY (CH): very stiff, high plasticity, dark grey, trace fine to coarse sand, trace fine gravel, moist.		0.18-0.70, B
		0.9		NATURAL Gravelly CLAY (CI): hard, medium plasticity, dark brown, fine to medium sized gravel, with fine to coarse sand, moist.		0.60-0.90, U50 PP>600
		1.0				
		1.5				
		1.6		Extremely weathered SILTSTONE (XW): recovered as Clayey Gravel - very dense, fine to medium grained, light brown, low, moist.		
		2.0				
		2.5		SILTSTONE (DW): distinctly weathered, low strength, light brown, fine to medium grained, moist.		
				BH04 Refusal at 2.7 m		

Comments

- Groundwater not encountered
- Borehole terminated at 'TC' bit refusal

Key

U - Undisturbed Tube
 D - Disturbed (small)
 B - Bulk Disturbed (large)
 ASS - Acid Sulfate Soil
 PP - Pocket Penetrometer
 VS - Vane Shear
 XW - Extremely Weathered
 DW - Distinctly Weathered
 Groundwater Observed
 Standing Water

Approved By: CJ

Approval Date: 14/01/2026

Job No : 1-29688
 Client : Thomas Adsett
 Project : CQU ROK Tafe Consolidation
 Location : Central Queensland University, Norman Gardens.

Easting : 246360.00
 Northing : 7418572.00
 RL : N/A
 Drill Rig : CDR-2

Sheet : 1 OF 1
 Logged : Rick Hayward
 Logged Date : 10/12/2025
 Checked : CJ

Drilling Method	Water	Depth (m)	Graphic Log	Material Description	DCP	Samples and Remarks
Hand Auger		0.13		NATURAL Silty CLAY (CH): stiff, high plasticity, dark brown, trace fine to medium sand, trace fine gravel, moist, with organics.		0.00-0.60, B
		0.3		NATURAL Silty CLAY (CH): stiff, high plasticity, dark brown, trace fine to medium sand, trace fine gravel, moist.		
		0.5		NATURAL Silty CLAY (CH): very stiff, high plasticity, dark grey, trace fine to medium sand, trace fine gravel, moist.		
		1.0				1.00-1.25, U50 PP230
		1.3		NATURAL Sandy CLAY (CL-CI): hard, low to medium plasticity, dark brown, fine to coarse grained sand, with fine to medium gravel, moist.		1.50-1.80, D
		1.5				
		2.0				
		2.5				
		2.6		Extremely weathered SILTSTONE (XW): recovered as Clayey Gravel - very dense, fine to medium grained, dark brown, low, moist.		2.60-3.00, D
		3.0				
		3.4		SILTSTONE (DW): distinctly weathered, low strength, dark brown, fine to medium grained, moist.		
		3.5				
				BH05 Refusal at 3.6 m		

Comments

- Groundwater not encountered
- Borehole terminated at 'TC' bit refusal

Key

U - Undisturbed Tube
 D - Disturbed (small)
 B - Bulk Disturbed (large)
 ASS - Acid Sulfate Soil
 PP - Pocket Penetrometer
 VS - Vane Shear
 XW - Extremely Weathered
 DW - Distinctly Weathered
 Groundwater Observed
 Standing Water

Approved By: CJ

Approval Date: 14/01/2026

Job No : 1-29688
 Client : Thomas Adsett
 Project : CQU ROK Tafe Consolidation
 Location : Central Queensland University, Norman Gardens.

Easting : 246256.00
 Northing : 7418438.00
 RL : N/A
 Drill Rig : CDR-2

Sheet : 1 OF 1
 Logged : Rick Hayward
 Logged Date : 10/12/2025
 Checked : CJ

Drilling Method	Water	Depth (m)	Graphic Log	Material Description	DCP	Samples and Remarks
Hand Auger		0.14		FILL Sandy Silty GRAVEL (GM): loose, fine to coarse sized, dark brown, fine to coarse grained sand, moist, trace of organics.		
		0.5		FILL Clayey SAND (SC): medium dense, fine to coarse grained, dark brown, low plasticity clay, with fine to medium gravel, moist.		
		1.0		NATURAL Silty CLAY (CH): very stiff, high plasticity, dark brown, trace fine to coarse sand, trace fine gravel, moist.		0.50-1.00, B
		1.5				0.80-1.10, U50 PP360
		1.6		NATURAL Sandy CLAY (CI): hard, medium plasticity, light brown, fine to medium grained sand, trace fine gravel, moist.		
		2.0				
		2.3		NATURAL Sandy CLAY (CL-CI): hard, low to medium plasticity, light brown, fine to medium grained sand, moist.		
		2.5				
		3.0				3.00-3.25, U50 PP>600
		3.5				
		3.6		NATURAL Sandy CLAY (CI): hard, medium plasticity, dark brown, fine to medium grained sand, trace fine to medium gravel, moist.		
		4.0				4.00-4.50, D
				BH06 Terminated at 4.5 m		

Comments

1. Groundwater not encountered

Key

U - Undisturbed Tube
 D - Disturbed (small)
 B - Bulk Disturbed (large)
 ASS - Acid Sulfate Soil
 PP - Pocket Penetrometer
 VS - Vane Shear
 XW - Extremely Weathered
 DW - Distinctly Weathered
 Groundwater Observed
 Standing Water

Approved By: CJ

Approval Date: 14/01/2026

Job No : 1-29688
 Client : Thomas Adsett
 Project : CQU ROK Tafe Consolidation
 Location : Central Queensland University, Norman Gardens.

Easting : 246221.00
 Northing : 7418381.00
 RL : N/A
 Drill Rig : CDR-2

Sheet : 1 OF 1
 Logged : Rick Hayward
 Logged Date : 10/12/2025
 Checked : CJ

Drilling Method	Water	Depth (m)	Graphic Log	Material Description	DCP	Samples and Remarks
Hand Auger		0.13		MULCH.		
		0.25		FILL Silty SAND (SM): very loose, fine to medium grained, dark brown, trace fine gravel, moist, trace of organics.		
		0.4		FILL Clayey Sandy GRAVEL (GC): medium dense, fine to medium sized, dark brown, fine to coarse grained sand, low plasticity clay, moist.		0.40-0.70, D
		0.8		FILL Sandy CLAY (CI-CH): very stiff, medium to high plasticity, dark brown, fine to coarse grained sand, with fine to medium gravel, moist.		1.00-1.40, D
		1.0		NATURAL Silty CLAY (CH): very stiff, high plasticity, dark brown, trace fine to coarse sand, trace fine gravel, moist.		1.20-1.45, U50 PP>600
		1.4		NATURAL Sandy CLAY (CL-CI): hard, low to medium plasticity, light brown, fine to coarse grained sand, trace fine gravel, moist.		
		1.5				
		2.0				
		2.5				
		3.0				3.00-3.30, U50 PP>600
		3.5				
		4.0				
		4.3				
		4.5		Extremely weathered SILTSTONE (XW): recovered as Gravelly Clay - hard, light brown, low to medium plasticity, fine to medium grained gravel, moist.		
BH07 Terminated at 5 m						

Comments

1. Groundwater not encountered

Key

U - Undisturbed Tube
 D - Disturbed (small)
 B - Bulk Disturbed (large)
 ASS - Acid Sulfate Soil
 PP - Pocket Penetrometer
 VS - Vane Shear
 XW - Extremely Weathered
 DW - Distinctly Weathered
 Groundwater Observed
 Standing Water

Approved By: CJ

Approval Date: 14/01/2026

Job No : 1-29688
 Client : Thomas Adsett
 Project : CQU ROK Tafe Consolidation
 Location : Central Queensland University, Norman Gardens.

Easting : 246479.00
 Northing : 7418311.00
 RL : N/A
 Drill Rig : CDR-2

Sheet : 1 OF 1
 Logged : Rick Hayward
 Logged Date : 10/12/2025
 Checked : CJ

Drilling Method	Water	Depth (m)	Graphic Log	Material Description	DCP	Samples and Remarks
Hand Auger		0.2		FILL Sandy Silty GRAVEL (GM): medium dense, fine to coarse sized, dark brown, fine to coarse grained sand, moist, with organics.		0.00-0.15, D
		0.35				
		0.5		FILL Sandy CLAY (CI): very stiff, medium plasticity, dark brown, fine to coarse grained sand, trace fine gravel, moist.		0.35-1.00, B
		1.0		NATURAL Silty CLAY (CH): very stiff, high plasticity, dark grey mottled dark brown, trace fine to coarse sand, trace fine gravel, moist.		0.35-0.60, D
		1.1				
		1.4		NATURAL Silty CLAY (CH): hard, high plasticity, dark brown mottled dark grey, trace fine to coarse sand, with fine to medium gravel, moist.		1.20-1.40, D
		1.5				
		2.0		NATURAL Sandy CLAY (CL): hard, low plasticity, light brown mottled light grey, fine to medium grained sand, trace fine gravel, moist.		1.50-1.75, U50 PP>600
		2.2				
		2.5		NATURAL Sandy CLAY (CI-CH): hard, medium to high plasticity, dark brown, fine to coarse grained sand, with fine to medium gravel, moist.		2.50-3.00, D
		3.0				
		3.5				
		3.7				
		4.0		NATURAL Gravelly CLAY (CI-CH): hard, medium to high plasticity, dark brown mottled light grey, fine to medium sized gravel, with fine to coarse sand, moist.		4.00-4.50, D
				BH08 Terminated at 4.5 m		

Comments

1. Groundwater not encountered

Key

U - Undisturbed Tube
 D - Disturbed (small)
 B - Bulk Disturbed (large)
 ASS - Acid Sulfate Soil
 PP - Pocket Penetrometer
 VS - Vane Shear
 XW - Extremely Weathered
 DW - Distinctly Weathered
 Groundwater Observed
 Standing Water

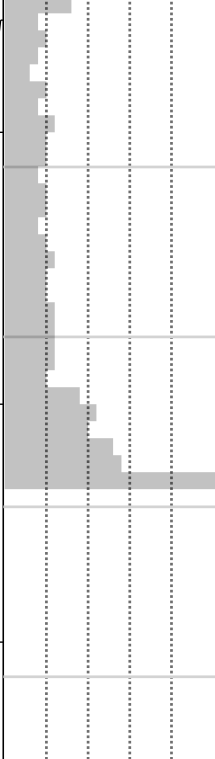
Approved By: CJ

Approval Date: 14/01/2026

Job No : 1-29688
 Client : Thomas Adsett
 Project : CQU ROK Tafe Consolidation
 Location : Central Queensland University, Norman Gardens.

Easting : 246508.00
 Northing : 7418298.00
 RL : N/A
 Drill Rig : CDR-2

Sheet : 1 OF 1
 Logged : Rick Hayward
 Logged Date : 10/12/2025
 Checked : CJ

Drilling Method	Water	Depth (m)	Graphic Log	Material Description	DCP	Samples and Remarks
Auger Drilling		0.14 0.5 0.8 1.0 1.5 2.0 2.4 2.5 3.0 3.5 3.8 4.0		FILL Clayey Sandy GRAVEL (GC): medium dense, fine to coarse sized, dark brown, fine to coarse grained sand, low to medium plasticity clay, moist, with organics. FILL Sandy CLAY (CI): very stiff, medium plasticity, dark brown, fine to coarse grained sand, trace fine to medium gravel, moist. NATURAL Silty CLAY (CH): very stiff, high plasticity, dark brown mottled dark grey, trace fine to coarse sand, trace fine gravel, moist. NATURAL Gravelly CLAY (CL-CI): hard, low to medium plasticity, light brown, fine to medium sized gravel, with fine to coarse sand, moist. Extremely weathered SILTSTONE (XW): recovered as Clayey Gravel - very dense, fine to medium grained, dark brown, low, moist.		1.00-1.30, U50 PP220
				BH09 Terminated at 4.5 m		

Comments

1. Groundwater not encountered

Key

U - Undisturbed Tube
 D - Disturbed (small)
 B - Bulk Disturbed (large)
 ASS - Acid Sulfate Soil
 PP - Pocket Penetrometer
 VS - Vane Shear
 XW - Extremely Weathered
 DW - Distinctly Weathered
 Groundwater Observed
 Standing Water

Approved By: CJ

Approval Date: 14/01/2026

Job No : 1-29688
 Client : Thomas Adsett
 Project : CQU ROK Tafe Consolidation
 Location : Central Queensland University, Norman Gardens.

Easting : 246705.00
 Northing : 7418708.00
 RL : N/A
 Drill Rig : CDR-2

Sheet : 1 OF 1
 Logged : Rick Hayward
 Logged Date : 09/12/2025
 Checked : CJ

Drilling Method	Water	Depth (m)	Graphic Log	Material Description	DCP	Samples and Remarks
Hand Auger		0.1				
		0.2		FILL Sandy Silty GRAVEL (GM): medium dense, fine to medium sized, dark brown, fine to coarse grained sand, moist.		
		0.5		FILL Clayey Sandy GRAVEL (GC): medium dense, fine to medium sized, dark brown, fine to coarse grained sand, low to medium plasticity clay, moist.		0.20-0.60, B
		0.6				0.20-0.60, D
		0.8		NATURAL Silty CLAY (CH): hard, high plasticity, dark brown, trace fine to coarse sand, trace fine gravel, moist, trace of root fibres.		0.80-1.20, D
		1.0		NATURAL Clayey GRAVEL (GC): medium dense to dense, fine to medium sized, dark brown, medium plasticity clay, trace fine to coarse sand, moist.		
		1.5		Extremely weathered SILTSTONE (XW): recovered as Gravelly Clay - hard, dark brown, medium plasticity, fine to medium grained gravel, moist.		
		2.0				
		2.1		As above: tending to clayey gravel.		
		2.5				
		3.0				3.00-3.15, U50 nil recovery crushed tube.
		3.5				
		4.0				
				BH10 Terminated at 4.5 m		

Comments

1. Groundwater not encountered

Key

U - Undisturbed Tube
 D - Disturbed (small)
 B - Bulk Disturbed (large)
 ASS - Acid Sulfate Soil
 PP - Pocket Penetrometer
 VS - Vane Shear
 XW - Extremely Weathered
 DW - Distinctly Weathered
 Groundwater Observed
 Standing Water

Approved By: CJ

Approval Date: 14/01/2026

Appendix C

Laboratory Test Certificates





**SOIL
SURVEYS**

Soil Surveys Engineering Pty Limited

Geotechnical Specialists

ABN: 70 054 043 631

www.soilsurveys.com.au

Helensvale Laboratory
Gold Coast
Unit 8, 140 Millaroo Drive
Helensvale
QLD, 4212
Tel: 07 3369 6000
info@soilsurveys.com.au

SHRINK SWELL INDEX REPORT

Client :	Thomas Adsett Architects	Report No:	1-29688 05.01.2026 ISS
Client Address :	C/O 1/87 Schneider Road, Eagle Farm, QLD, 4009	Report Date:	5/01/2026
Project Number :	1-29688	Issue Number:	1
Project :	CQU ROCK Tafe Consolidation	Page 1 of 1	
Location :	554/700 Yaamba Road, Norman Gardens		

SAMPLE DETAILS:

Sample ID:	1-29688-BH01-U00060	1-29688-BH02-U00100	1-29688-BH05-U00100	1-29688-BH06-U00080	1-29688-BH09-U00100
Date Sampled:	9/12/2025	9/12/2025	9/12/2025	9/12/2025	9/12/2025
Date Tested:	17/12/2025	17/12/2025	17/12/2025	17/12/2025	17/12/2025
Soil Description:	Silty Sandy Clay (CH) Trace of Gravel, Dark Brown	Silty Sandy Clay (CH) Trace of Gravel, Dark Brown	Silty Sandy Clay (CH) Trace of Gravel, Dark Brown	Silty Sandy Clay (CH) Trace of Gravel, Brown	Silty Sandy Clay (CI) Dark Brown
Sampling Method:	AS1726:2017 Geotechnical Site Investigations	AS1726:2017 Geotechnical Site Investigations	AS1726:2017 Geotechnical Site Investigations	AS1726:2017 Geotechnical Site Investigations	AS1726:2017 Geotechnical Site Investigations
Sampled by:	SSE - Driller RH	SSE - Driller RH	SSE - Driller RH	SSE - Driller RH	SSE - Driller RH
Sample Location:	BH01	BH02	BH05	BH06	BH09
Depth (m):	0.60-0.90	1.00-1.30	1.00-1.25	0.80-1.10	1.00-1.30
Source:	Borehole	Borehole	Borehole	Borehole	Borehole
Material:	Unknown	Unknown	Unknown	Unknown	Unknown

TEST RESULTS:

Swell Test (AS1289.7.1.1)					
Swell on Saturation(%)	2.8	3.9	1.2	2.0	0.0
Moisture Content Before(%)	22.5	34.0	32.3	26.4	27.8
Moisture Content After(%)	35.0	35.4	34.4	27.3	28.7
Shrink Test (AS1289.7.1.1)					
Shrink on Drying(%)	5.4	4.3	6.5	6.2	2.3
Shrinkage Moisture Content(%)	21.7	28.1	27.9	26.6	24.4
Est. Inert Material(%)	2	2	2	2	0
Crumbling	None	None	None	None	None
Cracking	Moderate	Extensive	Extensive	Slight	Extensive
Results:					
Shrink Swell Index - Iss (%)	3.8	3.5	3.9	4.0	1.3
*Swell Pressure (kPa):	120	250	60	120	20
Unit Weight (t/m ³):	1.87	1.90	1.87	1.99	1.90

Remarks :



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APPROVED SIGNATORY
Mark Rutten

Nata Accreditation number: 15301



Soil Surveys Engineering Pty Limited
Geotechnical Specialists
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Helensvale Laboratory
Gold Coast
Unit 8, 140 Millaroo Drive
Helensvale
QLD, 4212
Tel: 07 3369 6000
info@soilsurveys.com.au

CALIFORNIA BEARING RATIO REPORT

Client :	Thomas Adsett Architects	Report No:	1-29688-13.01.2026 CBR
Client Address :	C/O 1/87 Schneider Road, Eagle Farm QLD 4009	Report Date:	13/01/2026
Project Number :	1-29688	Issue Number:	1
Project :	CQU ROK Tafe Consolidation	Page 1 of 1	
Location :	554/700 Yaamba Road, Norman Gardens		

SAMPLE DETAILS:

Sample ID:	1-29688-BH04-B00018	1-29688-BH06-B00050	1-29688-BH08-B00030		
Field Sample ID:	-	-	-		
Date Sampled:	9/12/2025	9/12/2025	9/12/2025		
Soil Description:	Silty Clay (CH) Trace of Sand, Dark Brown	Silty Clay (CH) Trace of Sand, Dark Brown	Silty Clay (CH) Trace of Sand, Dark Brown		
Sampling Method:	AS1726:2017 Geotechnical Site Investigations	AS1726:2017 Geotechnical Site Investigations	AS1726:2017 Geotechnical Site Investigations		
Sampled by:	SSE - Driller RH	SSE - Driller RH	SSE - Driller RH		
Sample Location:	BH04	BH06	BH08		
Depth (m):	0.18 - 0.70	0.50 - 1.00	0.30 - 1.00		
Source:	Borehole	Borehole	Borehole		
Material:	Unknown	Unknown	Unknown		
Specification:	100% STD	100% STD	100% STD		

TEST RESULTS:

Moisture Content (AS1289.2.1.1)					
Moisture Content (%)	30.1	24.6	22.5		
Date Tested:	15/12/2025	15/12/2025	15/12/2025		
Maximum Dry Density - Standard (AS1289.5.1.1):					
Standard Maximum Dry Density (t/m ³):	1.66	1.73	1.82		
Standard Optimum Moisture Content (%):	25.5	24.0	18.8		
Oversize Sieve (mm):	19	19	19		
Oversize Material (%):	0	0	0		
Date Tested:	6/01/2026	6/01/2026	6/01/2026		
California Bearing Ratio (AS1289.6.1.1):					
CBR Value (%):	2.0	3.5	3.0		
CBR Penetration :	2.5mm	2.5mm	2.5mm		
Maximum Dry Density (t/m ³):	1.66	1.73	1.82		
Optimum Moisture Content (%):	25.5	24.0	18.8		
Dry Density Before Soaking (t/m ³):	1.66	1.73	1.81		
Density Ratio Before Soaking (%):	100.0	100.0	99.5		
Moisture Content before Soaking (%):	25.4	23.6	18.9		
Moisture Ratio before Soaking (%):	99.5	98.5	100.5		
Dry Density after Soaking (t/m ³):	1.59	1.69	1.76		
Density Ratio after Soaking (%):	96.0	97.5	96.5		
Swell (%):	4.5	2.5	3.0		
Moisture Content of Top 30mm (%):	30.7	24.7	23.5		
Moisture Content of Remaining Depth (%):	25.5	23.9	18.9		
Compactive Effort:	Standard	Standard	Standard		
Surcharge Mass (kg):	4.5	4.5	4.5		
Period of Soaking (Days):	4	4	4		
Oversize Material:	N/A	N/A	N/A		
Oversize Material (%):	0	0	0		
Date Tested:	12/01/2026	12/01/2026	12/01/2026		
CBR Total Curing Time (hr):	78	78	76		
Plasticity Determination:	Tactile Assessment	Tactile Assessment	Tactile Assessment		

Remarks :



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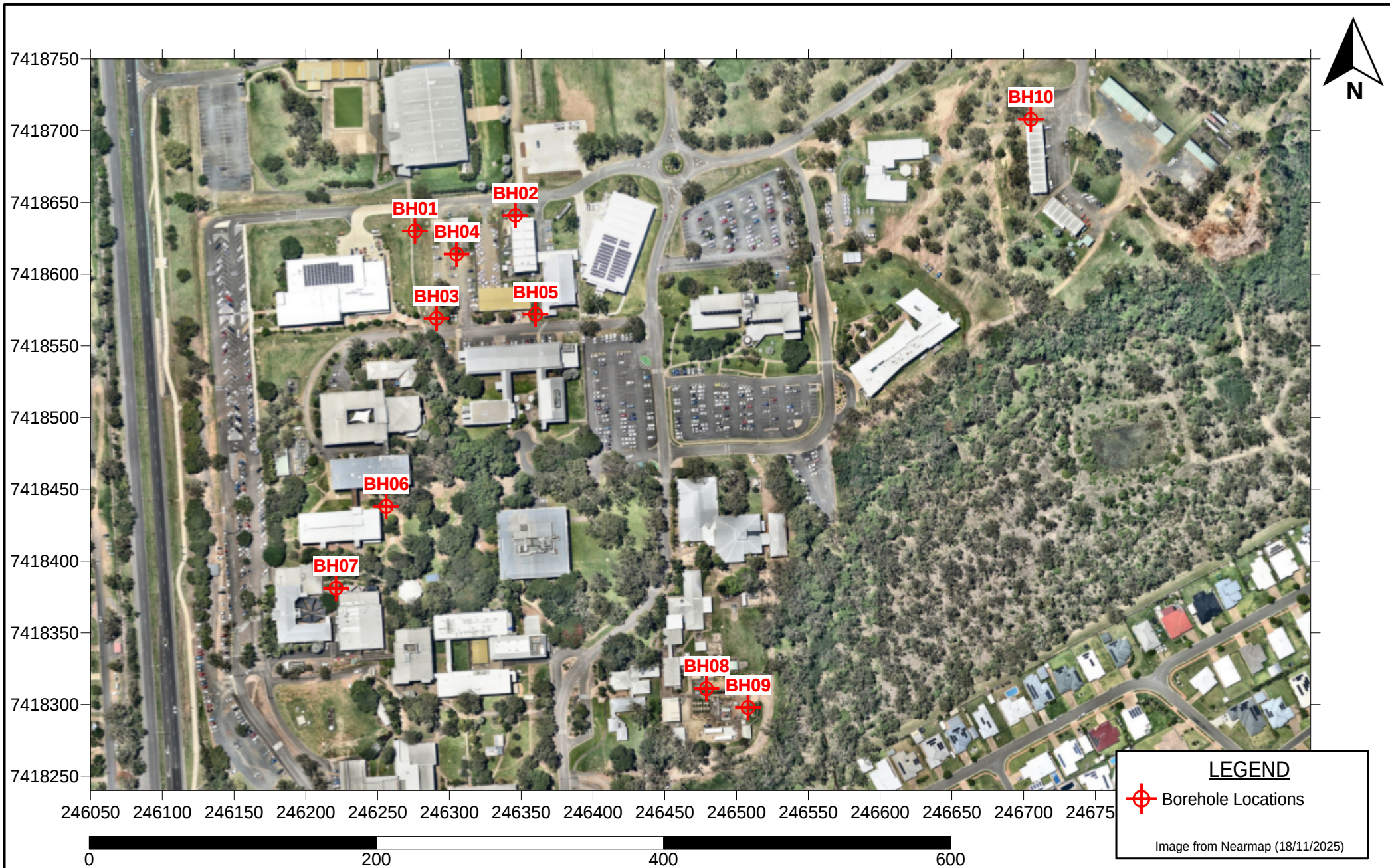
APPROVED SIGNATORY
Zane De Koster

NATA Accred No:15301

Appendix D

Site Plan





SCALE 1:3500 at A4			DRAWING TITLE BOREHOLE LOCATION PLAN CQU ROK Tafe Consolidation	CLIENT Thomas Adsett		 SOIL SURVEYS Geotechnical Specialists
DRAWING NUMBER 1-29688-01	DATE 14/01/2026	CHECKED CJ		LOCATION CQU, 554-700 Yaamba Road, Norman Gardens		